

## 1. Create one VPC, with 1 public subnet and 1 private subnet.

### Objective

To create an AWS VPC with one public subnet and one private subnet, configure an Internet Gateway, and configure separate route tables for public and private network traffic.

### Steps / Implementation

#### Step 1: Create VPC

1. Open the **AWS Management Console**.
2. Navigate to **VPC → Your VPCs**.
3. Click **Create VPC**.
4. Select **VPC only**.
5. Enter the following details:
  - a. VPC Name: TGW-Endpoint-VPC
  - b. IPv4 CIDR: 10.0.0.0/16
  - c. Tenancy: Default
6. Click **Create VPC**.

#### Step 2: Create Public Subnet

1. Navigate to **VPC → Subnets**.
2. Click **Create subnet**.
3. Select TGW-Endpoint-VPC.
4. Enter:
  - a. Subnet Name: TGW-Endpoint-Public-Subnet
  - b. Availability Zone: ap-south-1a
  - c. IPv4 CIDR: 10.0.1.0/24
5. Click **Create subnet**.

#### Step 3: Create Private Subnet

1. Click **Create subnet**.
2. Select TGW-Endpoint-VPC.
3. Enter:
  - a. Subnet Name: TGW-Endpoint-Private-Subnet
  - b. Availability Zone: ap-south-1a
  - c. IPv4 CIDR: 10.0.2.0/24
4. Click **Create subnet**.

#### Step 4: Create and Attach Internet Gateway

1. Navigate to **VPC → Internet Gateways**.
2. Click **Create internet gateway**.
3. Name it TGW-Endpoint-IGW.
4. Click **Create internet gateway**.
5. Select the Internet Gateway.
6. Choose **Actions → Attach to a VPC**.
7. Select TGW-Endpoint-VPC.
8. Click **Attach internet gateway**.

#### Step 5: Create Public Route Table

1. Navigate to **VPC → Route Tables**.
2. Click **Create route table**.
3. Enter:
  - a. Name: TGW-Endpoint-Public-RT
  - b. VPC: TGW-Endpoint-VPC
4. Click **Create route table**.
5. Open the **Routes** tab and select **Edit routes**.
6. Add the route:
  - a. Destination: 0.0.0.0/0
  - b. Target: TGW-Endpoint-IGW
7. Save the route.
8. Go to **Subnet associations → Edit subnet associations**.
9. Associate TGW-Endpoint-Public-Subnet.

#### Step 6: Create Private Route Table

1. Navigate to **Route Tables → Create route table**.
2. Enter:
  - a. Name: TGW-Endpoint-Private-RT
  - b. VPC: TGW-Endpoint-VPC
3. Click **Create route table**.
4. Go to **Subnet associations → Edit subnet associations**.
5. Associate TGW-Endpoint-Private-Subnet.
6. Verify that the route table contains the local route:

Destination: 10.0.0.0/16

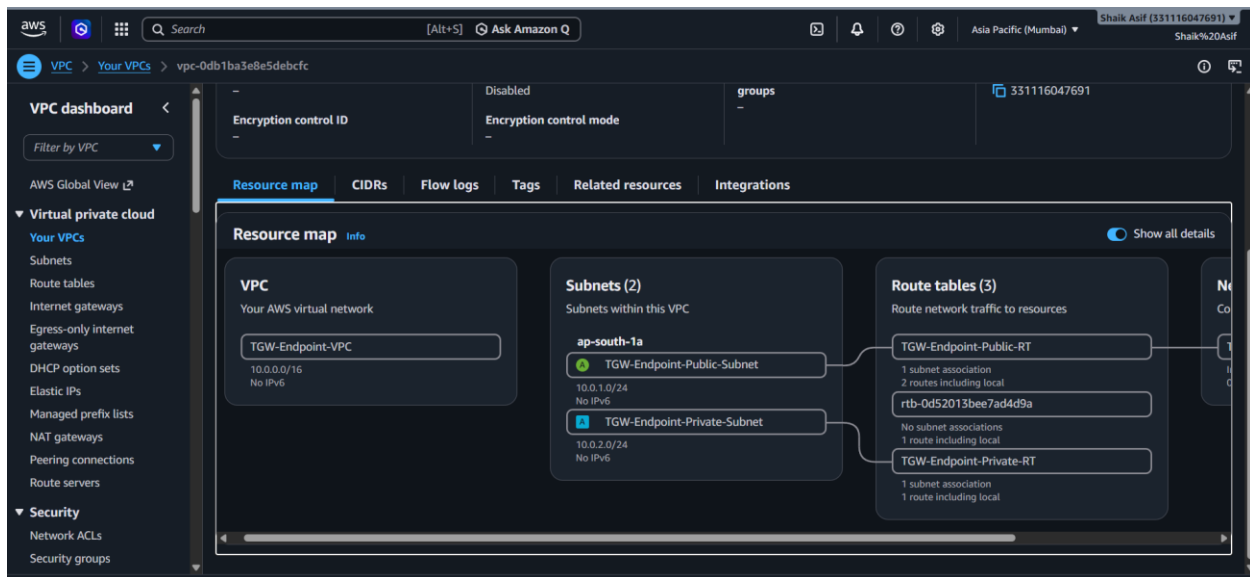
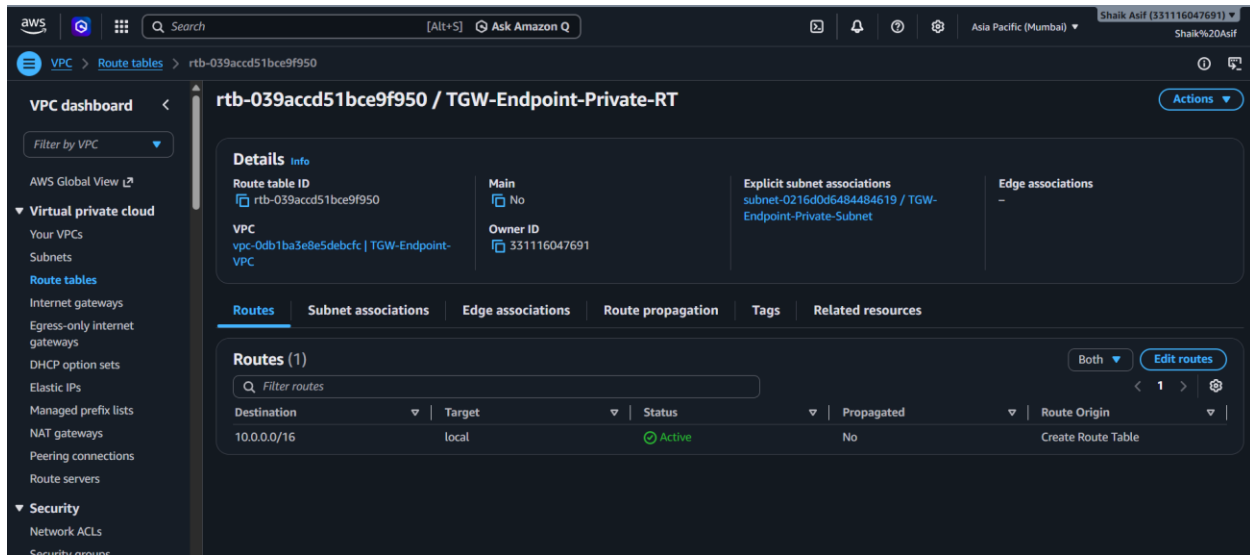
Target: local

7. No Internet Gateway route was added to the private route table.









## Result

Successfully created and configured TGW-Endpoint-VPC with CIDR block 10.0.0.0/16, one public subnet (10.0.1.0/24) and one private subnet (10.0.2.0/24). The Internet Gateway TGW-Endpoint-IGW was attached to the VPC. The public route table was configured with a default route to the Internet Gateway, while the private route table was configured with only the local VPC route. Both subnets were successfully associated with their respective route tables.

## 2. Enable VPC peering for cross-region.

## Objective

To establish a VPC peering connection between two VPCs located in different AWS Regions and configure routing to allow communication between the VPCs.

## Environment

| Resource              | Details                    |
|-----------------------|----------------------------|
| Requester VPC         | TGW-Endpoint-VPC           |
| Requester Region      | Mumbai (ap-south-1)        |
| Requester CIDR        | 10.0.0.0/16                |
| Accepter VPC          | TGW-Peering-VPC            |
| Accepter Region       | Singapore (ap-southeast-1) |
| Accepter CIDR         | 10.1.0.0/16                |
| Peering Connection ID | pcx-083900575e7926589      |

## Steps / Implementation

### Step 1: Create a VPC in a Different Region

1. Switched the AWS Region to Singapore (ap-southeast-1).
2. Navigated to VPC → Your VPCs.
3. Clicked Create VPC.
4. Selected VPC only.
5. Created the following VPC:
  - a. Name: TGW-Peering-VPC
  - b. IPv4 CIDR: 10.1.0.0/16
6. Clicked Create VPC.

### Step 2: Create Cross-Region VPC Peering

1. Switched back to Mumbai (ap-south-1).
2. Navigated to VPC → Peering connections.
3. Clicked Create peering connection.
4. Entered:
  - a. Name: TGW-Cross-Region-Peering

- b. Requester VPC: TGW-Endpoint-VPC
  - c. Acceptor Region: ap-southeast-1
  - d. Acceptor VPC ID: vpc-049c3dd264490e7e4
5. Clicked Create peering connection.
6. The peering request was created successfully and initially showed Pending Acceptance.

### Step 3: Accept the Peering Request

1. Switched to Singapore (ap-southeast-1).
2. Navigated to VPC → Peering connections.
3. Selected TGW-Cross-Region-Peering.
4. Selected Actions → Accept request.
5. Accepted the peering request.
6. Verified that the peering connection status changed to Active.

### Step 4: Configure Route in Mumbai

1. Switched to Mumbai (ap-south-1).
2. Navigated to VPC → Route Tables.
3. Selected TGW-Endpoint-Public-RT.
4. Opened Routes → Edit routes.
5. Added the following route:

| Destination | Target                |
|-------------|-----------------------|
| 10.1.0.0/16 | pcx-083900575e7926589 |

6. Saved the route.
7. Verified that the route status was Active..

### Step 5: Configure Route in Singapore

1. Switched to Singapore (ap-southeast-1).
2. Navigated to VPC → Route Tables.
3. Selected the route table associated with TGW-Peering-VPC.
4. Opened Routes → Edit routes.
5. Added the following route:

| Destination | Target                |
|-------------|-----------------------|
| 10.0.0.0/16 | pcx-083900575e7926589 |

6. Saved the route.
7. Verified that the route status was Active.

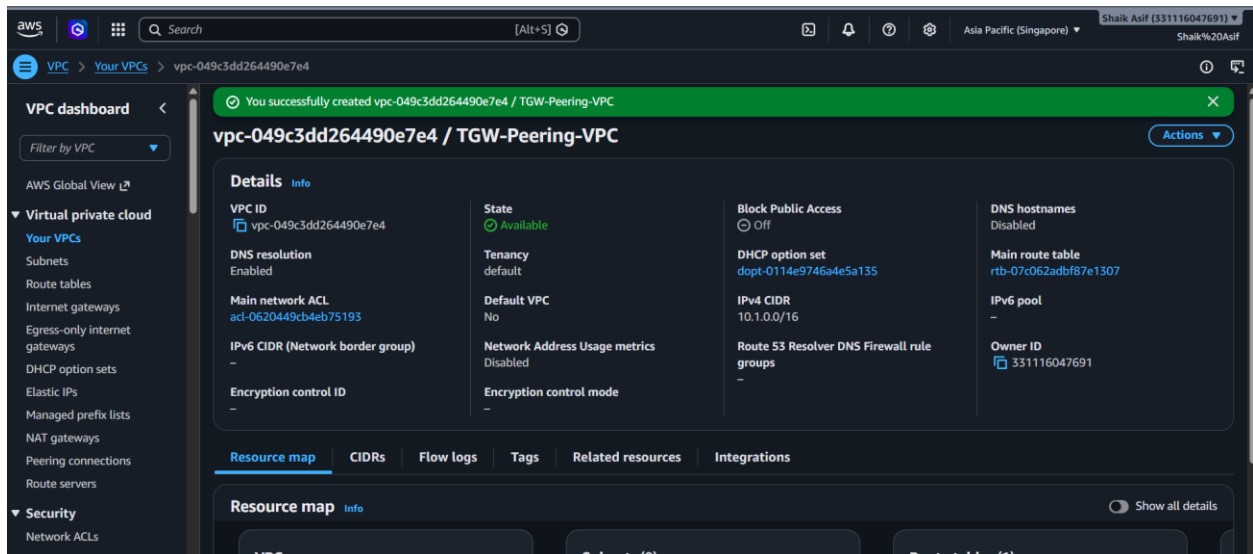
## Final Route Configuration

### Mumbai — TGW-Endpoint-VPC

10.0.0.0/16 → local  
10.1.0.0/16 → pcx-083900575e7926589  
0.0.0.0/0 → Internet Gateway

### Singapore — TGW-Peering-VPC

10.1.0.0/16 → local  
10.0.0.0/16 → pcx-083900575e7926589







## Result

Successfully established a cross-region VPC peering connection between TGW-Endpoint-VPC in Mumbai (ap-south-1) and TGW-Peering-VPC in Singapore (ap-southeast-1). The peering connection pcx-083900575e7926589 was successfully accepted and reached an Active state. Routes were configured in both VPC route tables to enable network communication between the two VPCs.

### **3. Enable VPC peering for cross-account (you can collaborate with your friend to do this task).**

## Objective

To establish a VPC peering connection between a VPC in my AWS account and a VPC in another AWS account, and configure the required routes for communication between the two VPCs.

## Environment

| Resource     | Details               |
|--------------|-----------------------|
| My VPC       | TGW-Endpoint-VPC      |
| My VPC CIDR  | 10.0.0.0/16           |
| My Region    | Mumbai (ap-south-1)   |
| Friend's VPC | vpc-02b14c4f0b3e7e2c4 |

|                     |                           |
|---------------------|---------------------------|
| Friend's VPC CIDR   | 10.2.0.0/16               |
| Friend's Region     | Oregon (us-west-2)        |
| Friend's Account ID | 634176595009              |
| Peering Type        | Cross-Account VPC Peering |

## Steps / Implementation

### Step 1: Prepare the Friend's VPC

1. Collaborated with a friend who has a separate AWS account.
2. The friend created a VPC in the Oregon (us-west-2) Region.
3. The VPC details were:
  - a. VPC ID: vpc-02b14c4f0b3e7e2c4
  - b. IPv4 CIDR: 10.2.0.0/16
4. Verified that the CIDR did not overlap with my VPC CIDR 10.0.0.0/16.

### Step 2: Create the Cross-Account VPC Peering Request

1. Logged in to my AWS account.
2. Switched to the Mumbai (ap-south-1) Region.
3. Navigated to VPC → Peering connections.
4. Clicked Create peering connection.
5. Entered the following details:
  - a. Name: TGW-Cross-Account-Peering
  - b. Requester VPC: TGW-Endpoint-VPC
  - c. Acceptor account: Another account
  - d. Acceptor Account ID: 634176595009
  - e. Acceptor Region: us-west-2
  - f. Acceptor VPC ID: vpc-02b14c4f0b3e7e2c4
6. Clicked Create peering connection.
7. The request was created with Pending Acceptance status.

### Step 3: Accept the Peering Request

1. The friend logged in to their AWS account.
2. Switched to the Oregon (us-west-2) Region.
3. Navigated to VPC → Peering connections.
4. Selected the pending peering request.
5. Selected Actions → Accept request.
6. Accepted the peering request.

7. Verified that the peering connection status changed to Active.

#### Step 4: Configure Route in My AWS Account

1. Logged in to my AWS account.
2. Navigated to VPC → Route Tables.
3. Selected TGW-Endpoint-Public-RT.
4. Opened Routes → Edit routes.
5. Added the following route:

| Destination | Target                    |
|-------------|---------------------------|
| 10.2.0.0/16 | Cross-Account VPC Peering |

6. Saved the route.
7. Verified that the route status was Active.

#### Step 5: Configure Route in Friend's AWS Account

1. The friend logged in to their AWS account.
2. Navigated to VPC → Route Tables.
3. Selected the route table associated with vpc-02b14c4f0b3e7e2c4.
4. Opened Routes → Edit routes.
5. Added the following route:

| Destination | Target                    |
|-------------|---------------------------|
| 10.0.0.0/16 | Cross-Account VPC Peering |

6. Saved the route.
7. Verified that the route status was Active.

#### Final Configuration

##### My AWS Account

VPC: 10.0.0.0/16

10.0.0.0/16 → local

10.2.0.0/16 → Cross-Account VPC Peering

## Friend's AWS Account

VPC: 10.2.0.0/16

10.2.0.0/16 → local

10.0.0.0/16 → Cross-Account VPC Peering

The screenshot shows the 'Create peering connection' page in the AWS console. The page is titled 'VPC > Peering connections > Create peering connection'. Below the title, there is a search bar and a 'Create a tag with a key of 'Name' and a value that you specify.' prompt. The main form is titled 'TGW-Cross-Account-Peering'. It has two main sections: 'Select a local VPC to peer with' and 'Select another VPC to peer with'. In the first section, 'VPC ID (Requester)' is set to 'vpc-0db1ba3e8e5debcfc (TGW-Endpoint-VPC)'. Below it, a table shows 'VPC CIDRs for vpc-0db1ba3e8e5debcfc (TGW-Endpoint-VPC)' with one entry: '10.0.0.0/16' with status 'Associated' and status reason '-'. The second section, 'Select another VPC to peer with', has 'Account' set to 'Another account' with 'Account ID' '634176595009'. 'Region' is set to 'Another Region' with 'United States (Oregon) (us-west-2)' selected.

Search [Alt+S]

VPC > Peering connections > Create peering connection

Create a tag with a key of 'Name' and a value that you specify.

TGW-Cross-Account-Peering

Select a local VPC to peer with

VPC ID (Requester)

vpc-0db1ba3e8e5debcfc (TGW-Endpoint-VPC)

VPC CIDRs for vpc-0db1ba3e8e5debcfc (TGW-Endpoint-VPC)

| CIDR        | Status     | Status reason |
|-------------|------------|---------------|
| 10.0.0.0/16 | Associated | -             |

Select another VPC to peer with

Account

☐ My account

☒ Another account

Account ID

634176595009

Region

☐ This Region (ap-south-1)

☒ Another Region

United States (Oregon) (us-west-2)

The screenshot shows the 'VPC dashboard' in the AWS console. The left sidebar contains a 'Virtual private cloud' section with links to 'Your VPCs', 'Subnets', 'Route tables', 'Internet gateways', 'Egress-only internet gateways', 'DHCP option sets', 'Elastic IPs', 'Managed prefix lists', 'NAT gateways', 'Peering connections', and 'Route servers'. The main content area is titled 'pcx-057b9bb85e6616c65 / TGW-Cross-Account-Peering'. It has a 'Details' tab selected, showing 'Requester owner ID' (331116047691), 'Peering connection ID' (pcx-057b9bb85e6616c65), 'Status' (Active), and 'Expiration time' (-). It also shows 'Accepter owner ID' (634176595009), 'Requester VPC' (vpc-0db1ba3e8e5debcfc / TGW-Endpoint-VPC), 'Requester CIDRs' (10.0.0.0/16), 'Requester Region' (Mumbai (ap-south-1)), 'VPC Peering connection ARN' (arn:aws:ec2:ap-south-1:331116047691:vpc-peering-connection/pcx-057b9bb85e6616c65), 'Accepter VPC' (vpc-02b14c4f0b3e7e2c4), 'Accepter CIDRs' (10.2.0.0/16), and 'Accepter Region' (Oregon (us-west-2)). Below the details, there is a 'DNS settings' section with a 'Requester VPC' (vpc-0db1ba3e8e5debcfc / TGW-Endpoint-VPC) and a 'DNS settings' button. The settings show 'Allow accepter VPC to resolve DNS of hosts in requester VPC to private IP addresses' as 'Disabled'.

Search [Alt+S]

VPC > Peering connections > pcx-057b9bb85e6616c65

VPC dashboard

Filter by VPC

AWS Global View

Virtual private cloud

- Your VPCs
- Subnets
- Route tables
- Internet gateways
- Egress-only internet gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists
- NAT gateways
- Peering connections
- Route servers

Security

- Network ACLs
- Security groups

pcx-057b9bb85e6616c65 / TGW-Cross-Account-Peering

Actions

Details

Requester owner ID

331116047691

Peering connection ID

pcx-057b9bb85e6616c65

Status

Active

Expiration time

-

Accepter owner ID

634176595009

Requester VPC

vpc-0db1ba3e8e5debcfc / TGW-Endpoint-VPC

Requester CIDRs

10.0.0.0/16

Requester Region

Mumbai (ap-south-1)

VPC Peering connection ARN

arn:aws:ec2:ap-south-1:331116047691:vpc-peering-connection/pcx-057b9bb85e6616c65

Accepter VPC

vpc-02b14c4f0b3e7e2c4

Accepter CIDRs

10.2.0.0/16

Accepter Region

Oregon (us-west-2)

DNS settings

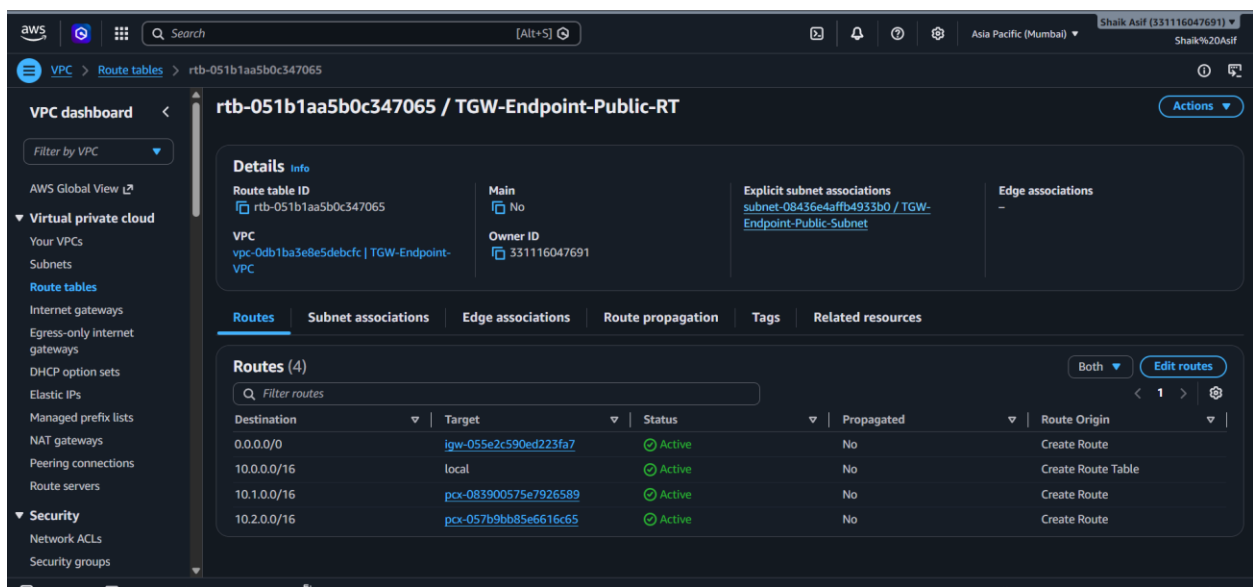
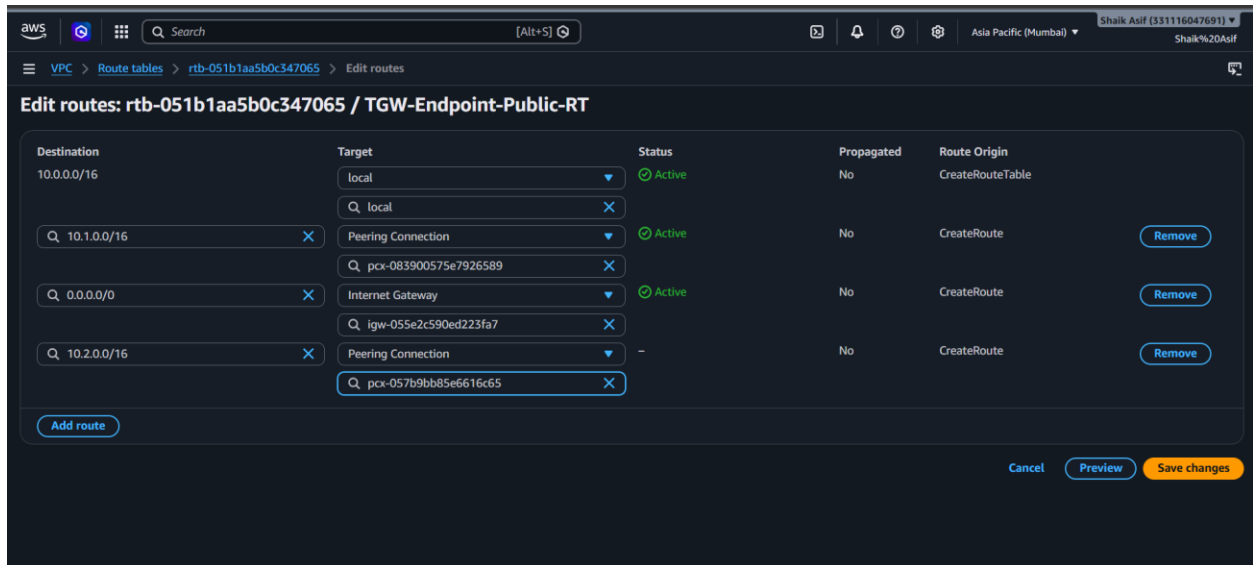
Requester VPC (vpc-0db1ba3e8e5debcfc / TGW-Endpoint-VPC)

Allow accepter VPC to resolve DNS of hosts in requester VPC to private IP addresses

Disabled

Accepter VPC (vpc-02b14c4f0b3e7e2c4)

Edit DNS settings



## Result

Successfully established a cross-account VPC peering connection between TGW-Endpoint-VPC in my AWS account and vpc-02b14c4f0b3e7e2c4 in my friend's AWS account. The VPCs were located in different AWS Regions, with CIDR blocks 10.0.0.0/16 and 10.2.0.0/16. The peering request was accepted by the friend and became Active. Routes were configured in both accounts to enable communication between the two VPC networks.

## 4 Set up a VPC Transit Gateway

### Objective

To create an AWS Transit Gateway and attach the existing VPC to it, configure the Transit Gateway route table association and route propagation, and verify the VPC route through the Transit Gateway.

### Environment

| Resource        | Details                     |
|-----------------|-----------------------------|
| VPC             | TGW-Endpoint-VPC            |
| VPC CIDR        | 10.0.0.0/16                 |
| Region          | Mumbai (ap-south-1)         |
| Transit Gateway | TGW-DevOps                  |
| VPC Attachment  | TGW-Endpoint-VPC-Attachment |

### Steps / Implementation

#### Step 1: Create Transit Gateway

1. Logged in to the AWS Management Console.
2. Navigated to VPC → Transit Gateways.
3. Clicked Create transit gateway.

4. Entered the following details:
  - a. Name tag: TGW-DevOps
  - b. Description: Transit Gateway for DevOps VPC
  - c. DNS support: Enabled
  - d. VPN ECMP support: Enabled
  - e. Default route table association: Enabled
  - f. Default route table propagation: Enabled
5. Left the remaining settings as default.
6. Clicked Create transit gateway.
7. Waited until the Transit Gateway state changed to Available.

## Step 2: Create VPC Transit Gateway Attachment

1. Navigated to VPC → Transit Gateway Attachments.
2. Clicked Create transit gateway attachment.
3. Selected:
  - a. Name: TGW-Endpoint-VPC-Attachment
  - b. Transit Gateway: TGW-DevOps
  - c. Attachment type: VPC
  - d. VPC: TGW-Endpoint-VPC
4. Selected TGW-Endpoint-Public-Subnet as the subnet for the Transit Gateway attachment.
5. Clicked Create transit gateway attachment.
6. Waited until the attachment state changed to Available.

## Step 3: Verify Transit Gateway Route Table Association

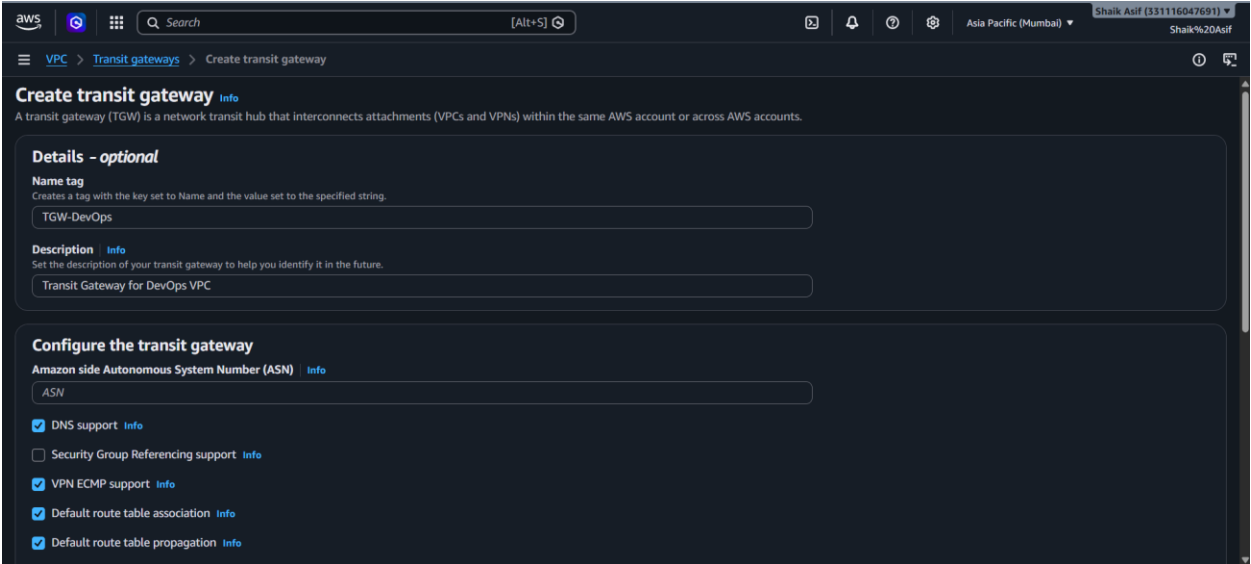
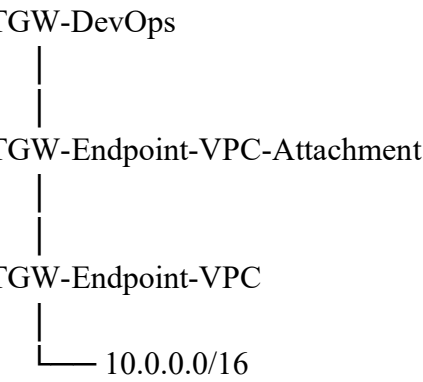
1. Navigated to VPC → Transit Gateway Route Tables.
2. Selected the route table associated with TGW-DevOps.
3. Opened the Associations tab.
4. Verified that TGW-Endpoint-VPC-Attachment was associated with the Transit Gateway route table.

## Step 4: Verify Route Propagation

1. Opened the Propagation tab of the Transit Gateway route table.
2. Verified that TGW-Endpoint-VPC-Attachment was configured for route propagation.
3. Opened the Routes tab.
4. Verified the propagated route:

| Destination | Target                      | Status |
|-------------|-----------------------------|--------|
| 10.0.0.0/16 | TGW-Endpoint-VPC-Attachment | Active |

Final Configuration



Site-to-Site VPN connections

Client VPN endpoints

AWS Verified Access

Verified Access instances

Verified Access trust providers

Verified Access groups

Verified Access endpoints

Transit gateways

Transit gateways

Transit gateway attachments

Transit gateway policy tables

Transit gateway route tables

Transit gateway multicast

Metering policies

Traffic Mirroring

tgw-0f62cce42b85749c3

tgw-0f62cce42b85749c3 / TGV-DevOps

Actions

Details

Transit gateway ID

tgw-0f62cce42b85749c3

State

Available

Amazon ASN

64512

DNS support

Enable

Transit gateway ARN

arn:aws:ec2:ap-south-1:331116047691:1:transit-gateway/tgw-0f62cce42b85749c3

Default association route table

Enable

Association route table ID

tgw-rtb-0a04d79e1f5bd3fbd

Auto accept shared attachments

Disable

Owner ID

331116047691

Default propagation route table

Enable

Propagation route table ID

tgw-rtb-0a04d79e1f5bd3fbd

VPN ECMP support

Enable

Description

Transit Gateway for DevOps VPC

Transit gateway CIDR blocks

-

Security Group Referencing support

Disable

Metering policy ID

-

Multicast support

Disable

Encryption support

Disabled

Encryption state message

-

Flow logs

Sharing

Tags

Flow logs

Find flow log by attribute or tag

Actions

Create flow log

CloudShell

Agent Toolkit for AWS

Feedback

Console Mobile App

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VPC

Transit gateway attachments

Create transit gateway attachment

Shaik Asif (331116047691)

Shaik%20Asif

Create transit gateway attachment

A transit gateway (TGW) is a network transit hub that interconnects attachments (VPCs and VPNs) within the same AWS account or across AWS accounts.

Details

Name tag - optional

Creates a tag with the key set to Name and the value set to the specified string.

Transit gateway ID

tgw-0f62cce42b85749c3

Attachment type

VPC

VPC attachment

Select and configure your VPC attachment.

☒ DNS support

☒ Security Group Referencing support

☐ IPv6 support

☐ Appliance Mode support

VPC ID

Select the VPC to attach to the transit gateway.

Search

[Alt+S]

Asia Pacific (Mumbai)

Shaik Asif (33116047691)

VPC > Transit gateway attachments > tgw-attach-039b9e808320208ad

VPC dashboard

Filter by VPC

AWS Global View

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Security

Network ACLs

tgw-attach-039b9e808320208ad / TGW-Endpoint-VPC-Attachment

Info

Actions

Details

Transit gateway attachment ID

tgw-attach-039b9e808320208ad

State

Available

Resource type

VPC

Association state

Associated

Transit gateway ID

tgw-0f62cce42b85749c3

Resource owner ID

331116047691

Resource ID

vpc-0db1ba3e8e5deb9cf

Association route table ID

tgw-rtb-0a04d79e1f5bd3fbd

Association policy table ID

-

Transit gateway owner ID

331116047691

DNS support

Enable

Security Group Referencing support

Enable

IPv6 support

Disable

Appliance Mode support

Disable

Subnet IDs

subnet-08436e4affb4933b0

Flow logs

Tags

Flow logs

Info

Find flow log by attribute or tag

Actions

Create flow log

Name

Flow log ID

Destination type

Destination name

IAM role ARN

No flow logs

Search

[Alt+S]

Asia Pacific (Mumbai)

Shaik Asif (33116047691)

VPC > Transit gateway route tables > tgw-rtb-0a04d79e1f5bd3fbd

Site-to-Site VPN connections

Client VPN endpoints

AWS Verified Access

Verified Access instances

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Verified Access groups

Verified Access endpoints

Transit gateways

Transit gateways

Transit gateway attachments

Transit gateway policy tables

Transit gateway route tables

Transit gateway multicast

Metering policies

Traffic Mirroring

tgw-rtb-0a04d79e1f5bd3fbd

Info

Actions

Details

Transit gateway route table ID

tgw-rtb-0a04d79e1f5bd3fbd

State

Available

Default association route table

Yes

Default propagation route table

Yes

Transit gateway ID

tgw-0f62cce42b85749c3

Associations

Propagations

Prefix list references

Routes

Tags

Propagations (1)

Info

Find propagation by attribute or tag

Delete propagation

Create propagation

Attachment ID

Resource type

Resource ID

State

tgw-attach-039b9e808320208ad

VPC

vpc-0db1ba3e8e5deb9cf

Enabled

Search

[Alt+S]

Asia Pacific (Mumbai)

Shaik Asif (33116047691)

VPC > Transit gateway route tables > tgw-rtb-0a04d79e1f5bd3fbd

Site-to-Site VPN connections

Client VPN endpoints

AWS Verified Access

Verified Access instances

Verified Access trust providers

Verified Access groups

Verified Access endpoints

Transit gateways

Transit gateways

Transit gateway attachments

Transit gateway policy tables

Transit gateway route tables

Transit gateway multicast

Metering policies

Traffic Mirroring

tgw-rtb-0a04d79e1f5bd3fbd

Info

Actions

Details

Transit gateway route table ID

tgw-rtb-0a04d79e1f5bd3fbd

State

Available

Default association route table

Yes

Default propagation route table

Yes

Transit gateway ID

tgw-0f62cce42b85749c3

Associations

Propagations

Prefix list references

Routes

Tags

Filter routes by CIDR (2)

Exact CIDR

Select a valid IP4 or IPv6 CIDR.

0.0.0.0/0, ::/0

Longest prefix match

Enter a valid IP4 or IPv6 and press enter.

0.0.0.0, ::

Supernet of match

Select a valid IP4 or IPv6 CIDR.

0.0.0.0/0, ::/0

Subnet of match

Select a valid IP4 or IPv6 CIDR.

0.0.0.0/0, ::/0

Routes (1)

Info

Find route by attribute or tag

Actions

Create static route

CIDR

Attachment ID

Resource ID

Resource type

Route type

Route state

P.

10.0.0.0/16

tgw-attach-039b9e808320208ad

vpc-0db1ba3e8e5deb9cf

VPC

Propagated

Active

-

## Result

Successfully created the TGW-DevOps Transit Gateway and attached TGW-Endpoint-VPC using TGW-Endpoint-VPC-Attachment. The attachment reached an Available state, and the Transit Gateway route table association and route propagation were successfully configured. The VPC CIDR 10.0.0.0/16 was automatically propagated to the Transit Gateway route table and shown as an Active route.

## 5. Set up a VPC Endpoint.

### Objective

To create an Amazon S3 Gateway VPC Endpoint and associate it with the private route table so that resources in the private subnet can access Amazon S3 without using an Internet Gateway.

### Environment

| Resource      | Details                 |
|---------------|-------------------------|
| VPC           | TGW-Endpoint-VPC        |
| VPC CIDR      | 10.0.0.0/16             |
| Region        | Mumbai (ap-south-1)     |
| VPC Endpoint  | TGW-S3-VPC-Endpoint     |
| Service       | Amazon S3               |
| Endpoint Type | Gateway                 |
| Route Table   | TGW-Endpoint-Private-RT |

### Steps / Implementation

#### Step 1: Open VPC Endpoints

1. Logged in to the AWS Management Console.
2. Navigated to VPC → Endpoints.
3. Selected the Mumbai (ap-south-1) Region.
4. Clicked Create endpoint.

## Step 2: Select the S3 Service

1. Entered the endpoint name:  
TGW-S3-VPC-Endpoint
2. Selected AWS services.
3. Selected the S3 service:  
com.amazonaws.ap-south-1.s3
4. Selected the Gateway endpoint type.
5. Did not enable the Cross-Region endpoint option.

Evidence: Screenshot showing the S3 Gateway service selected.

## Step 3: Select the VPC

1. Under Network settings, selected:  
TGW-Endpoint-VPC
2. Verified the VPC CIDR:  
10.0.0.0/16

Evidence: Screenshot showing the selected VPC.

## Step 4: Associate the Private Route Table

1. Selected the route table:  
TGW-Endpoint-Private-RT
2. Created the VPC Endpoint.
3. After creation, verified that the endpoint was associated with the private route table.

Evidence: Screenshot showing the endpoint and associated route table.

## Step 5: Verify the Route Table

1. Navigated to VPC → Route Tables.
2. Selected:  
TGW-Endpoint-Private-RT
3. Opened the Routes tab.
4. Verified the local VPC route:  
10.0.0.0/16 → local
5. Verified that the S3 Gateway Endpoint added an S3 prefix-list route:

pl-xxxxxxxx → vpce-xxxxxxxx

The exact prefix-list and endpoint IDs depend on the AWS account.

Evidence: Screenshot of the private route table showing the S3 endpoint route.

## Final Configuration

TGW-Endpoint-VPC

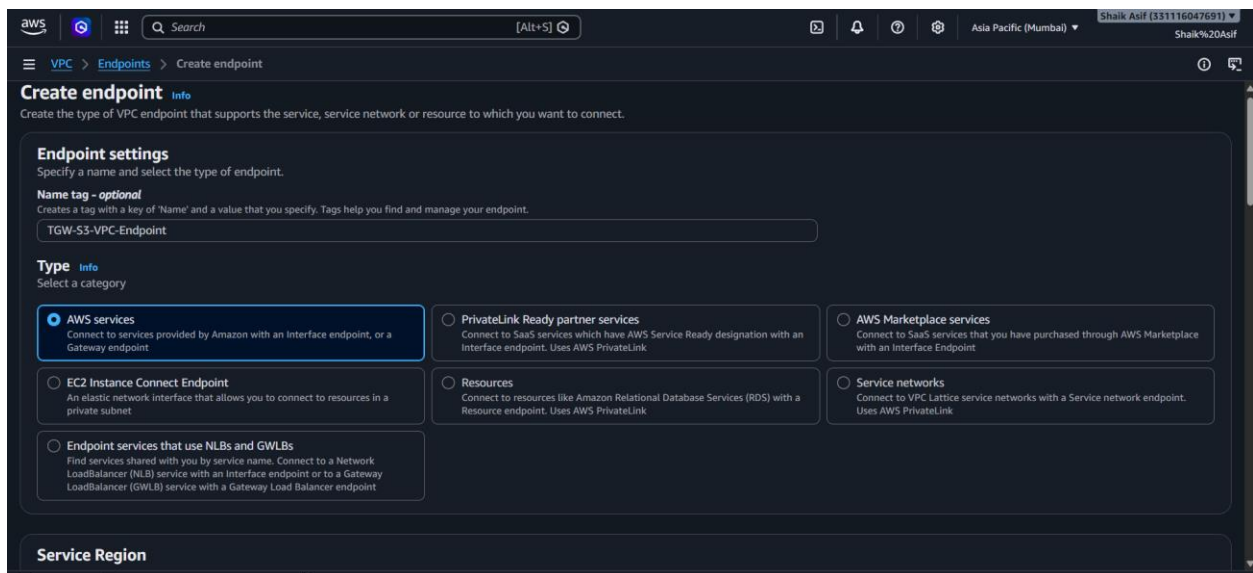
TGW-Endpoint-Private-Subnet

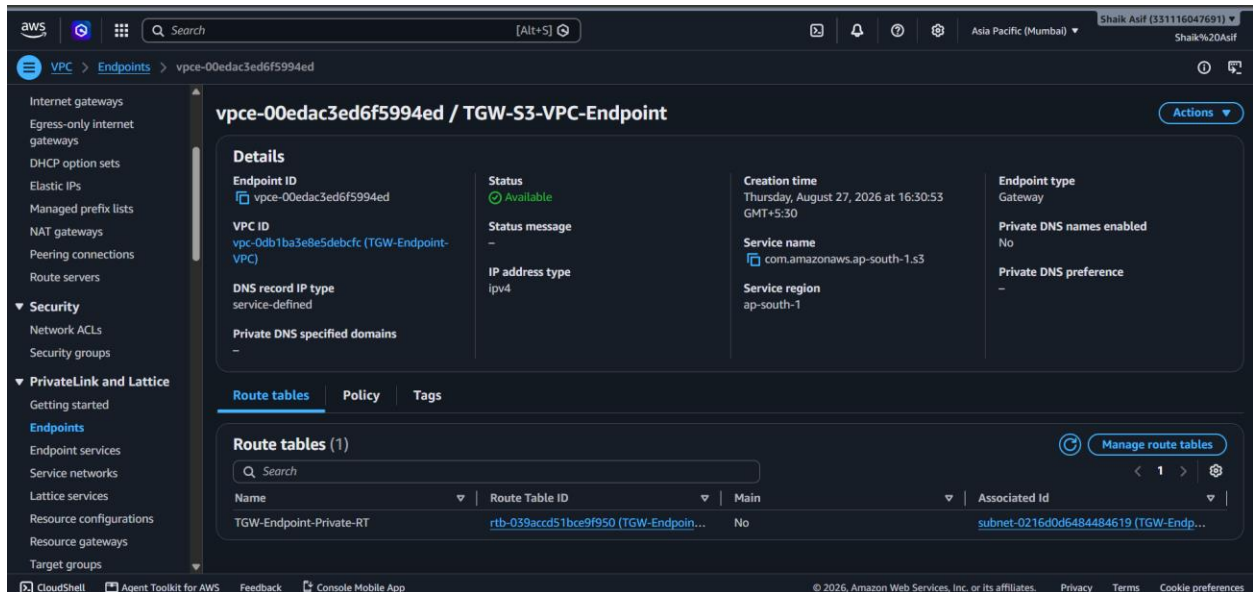
TGW-Endpoint-Private-RT

10.0.0.0/16 → local

S3 Prefix List → VPC Endpoint

Amazon S3





## Result

Successfully created the **Amazon S3 Gateway VPC Endpoint** named **TGW-S3-VPC-Endpoint** in the **TGW-Endpoint-VPC**. The endpoint was created in the **Mumbai (ap-south-1)** Region and associated with **TGW-Endpoint-Private-RT**. The **S3 prefix-list** route was added to the private route table, allowing resources in the private subnet to access Amazon S3 through the VPC Endpoint without requiring an Internet Gateway.